

3.1 Algebra

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Advanced algebra includes factorisation, partial fractions and binomial series as appropriate. For a rational expression, divide first if numerator degree is not lower than denominator, then decompose using factors and repeated-factor terms. Validate by recombining or testing convenient x .

Worked example: follow the reasoning

$1/[x(x+1)] = 1/x - 1/(x+1)$, readily integrated as $\ln|x| - \ln|x+1| + C$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Decompose $3/[x(x+3)]$ into simple fractions.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Set $A/x + B/(x+3)$: $3 = A(x+3) + Bx$ [1]; $x=0$ gives $A=1$ [1], $x=-3$ gives $B=-1$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.